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ORBITAL TOOLS

Abstract

An orbital tool operable in first and second rotational directions, and includes a sprag clutch operably coupled to an output shaft to the tool. When the tool is operated in the first rotational direction, the sprag clutch allows free spinning of an output shaft, thus providing operation in a random orbital mode. When the tool is operated in the second rotational direction, the sprag clutch locks or prevents the output shaft from free spinning, thus providing operation in a rotary mode.

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Background/Summary

TECHNICAL FIELD OF THE INVENTION
[0001] The present invention relates generally to orbital tools, and more particularly, to orbital tools that operate in different modes based on direction of rotation.

BACKGROUND OF THE INVENTION

[0002] Orbital tools, such as, for example, orbital sanders and/or polishers, are commonly used in automotive, industrial, and household applications to remove material from, smooth, and/or polish a surface. A random orbital sander is one type of orbital tool that operates in a random-orbit action by simultaneously spinning a sanding disk and moving it in overlapping ellipses, while also allowing the sanding disk to free spin. A rotary orbital sander is another type of orbital tool, similar to a random orbital sander. However, the rotary orbital sander operates without allowing the sanding disk to free spin. Therefore, traditionally, multiple different types of tools are required to perform a random orbital polisher/sander related task and a rotary polisher/sander related task.

SUMMARY OF THE INVENTION

[0003] The present invention relates broadly to an orbital tool that is selectively operable in first and second rotational directions. The tool includes a sprag clutch (also known as a one-way bearing, freewheel clutch, cam clutch, or overrunning clutch) operably coupled to an output shaft of the tool. When the tool is operated in the first rotational direction, the sprag clutch allows free spinning of an output shaft, thus providing operation in a random orbital mode. When the tool is operated in the second rotational direction, the sprag clutch locks or prevents the output shaft from free spinning, thus providing operation in a rotary mode.

[0004] In an embodiment, the present invention includes an orbital tool including a tool housing having a motor housing portion with a motor housing longitudinal axis and a handle housing portion with a handle housing longitudinal axis, wherein the handle housing longitudinal axis is disposed at an angle of about 90 degrees to 120 degrees relative to the motor housing longitudinal axis. Thus, the tool housing forms a pistol-type grip tool. A motor is disposed in the motor housing portion and includes a motor shaft, wherein the motor is adapted to selectively rotate the motor shaft in either of first and second rotational directions. A balancer may be operably coupled to the motor shaft, and an output shaft is disposed in the balancer. A sprag clutch is also disposed in the balancer and operably coupled to the output shaft, wherein the sprag clutch is adapted to allow rotation of the output shaft relative to the balancer when the motor shaft is selectively rotated in the first rotational direction, and prevent rotation of the output shaft relative to the balancer when the motor shaft is selectively rotated in the second rotational direction.

[0005] In another embodiment, the present invention includes an orbital tool including a tool housing, a nose housing including an illumination element and that is coupled to the tool housing, and a motor disposed in the tool housing and including a motor shaft, wherein the motor is adapted to selectively rotate the motor shaft in either of first and second rotational directions. A balancer is operably coupled to the motor shaft, an output shaft is disposed in the balancer, and a sprag clutch is disposed in the balancer and operably coupled to the output shaft, wherein the sprag clutch is adapted to allow rotation of the output shaft relative to the balancer when the motor shaft is rotated in the first rotational direction, and prevent rotation of the output shaft with respect to the balancer when the motor shaft is rotated in the second rotational direction.

[0006] In another embodiment, the present invention includes an orbital tool including a housing and a motor disposed in the housing and including a motor shaft, wherein the motor is adapted to selectively rotate the motor shaft in either of first and second rotational directions. A balancer is operably coupled to the motor shaft, a counterbalance is coupled to an exterior surface of the balancer, an output shaft is disposed in the balancer, and a sprag clutch is disposed in the balancer and operably coupled to the output shaft. The sprag clutch is adapted to allow rotation of the output shaft relative to the balancer when the motor shaft is rotated in the first rotational direction, and prevent rotation of the output shaft relative to the balancer when the motor shaft is rotated in the second rotational direction.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0007] For the purpose of facilitating an understanding of the subject matter sought to be protected, there is illustrated in the accompanying drawing embodiments thereof, from an inspection of which, when considered in connection with the following description, the subject matter sought to be protected, its construction and operation, and many of its advantages, should be readily understood and appreciated.

[0008] FIG. **1** is perspective view of an exemplar orbital tool, incorporating an embodiment of the present invention.

[0009] FIG. **2** is a first side view of the orbital tool of FIG. **1**.

[0010] FIG. **3** is a second side view of the orbital tool of FIG. **1**.

[0011] FIG. **4** is a partial exploded, perspective view of the orbital tool of FIG. **1**.

[0012] FIG. **5** is an exploded, perspective view of the orbital tool of FIG. **1**.

[0013] FIG. **6** is a perspective view of internal components of the orbital tool of FIG. **1**.

[0014] FIG. **7** is an exploded perspective view of components of the orbital tool of FIG. **1**.

[0015] FIG. **8** is an exploded perspective view of output components of the orbital tool of FIG. **1**.

[0016] FIG. **9** is a cross sectional view of the output components of the orbital tool of FIG. **8** assembled together.

[0017] FIG. **10** is a cross sectional view of another embodiment of output components of the orbital tool, according to an embodiment of the present invention.

[0018] FIG. **11** is a cross sectional view of another embodiment of output components of the orbital tool, according to an embodiment of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0019] While the present invention is susceptible of embodiments in many different forms, there is shown in the drawings, and will herein be described in detail, a preferred embodiment of the invention with the understanding that the present disclosure is to be considered as an exemplification of the principles of the invention and is not intended to limit the broad aspect of the invention to embodiments illustrated. As used herein, the term “present invention” is not intended to limit the scope of the claimed invention and is instead a term used to discuss exemplary embodiments of the invention for explanatory purposes only.

[0020] The present invention relates broadly to an orbital tool that is selectively operable in first and second rotational directions. The tool includes a sprag clutch (also known as a one-way bearing, freewheel clutch, cam clutch, or overrunning clutch) operably coupled to an output shaft of the tool. When the tool is operated in the first rotational direction, the sprag clutch allows free spinning of an output shaft, thus providing operation in a random orbital mode. When the tool is operated in the second rotational direction, the sprag clutch locks or prevents the output shaft from free spinning, thus providing operation in a rotary mode.

[0021] Referring to FIGS. **1-6**, an exemplary tool **100**, such as an orbital sander/polisher sander, incorporating an embodiment of the present invention is illustrated. The tool **100** includes a tool housing **102**, an output assembly **104** operably coupled to the tool housing **102**, a motor **106** disposed in the tool housing **102** and operably coupled to the output assembly **104**, a trigger **108** operably coupled to the motor **106** and actuatable to operate the motor **106** and thereby the output assembly **104**, and a direction selector switch **110** adapted to select the rotational direction of the motor **106**. The tool housing **102** includes a motor housing portion **112** and a handle housing portion **114**. The motor housing portion **112** is adapted to house the motor **106**. The handle housing portion **114** extends from the motor housing portion **112**. As illustrated, the tool **100** is a pistol grip style tool, and the motor housing portion **112** and handle housing portion **114** may be disposed at an angle relative to each other. For example, a motor housing longitudinal axis of the motor housing portion **112** and a handle housing longitudinal axis of the handle housing portion **114** may be disposed at an angle of about 90 to about 120 degrees, and more particularly about 110 degrees, relative to each other. In an example, the tool housing **102** may include first and second housing portions **116**, **118** (respectively forming first and second sides of the housing **102**) that are coupled together in a clamshell type manner and coupled to the output assembly **104**. In another example, the tool housing **102** (including the first and second housing portions **116**, **118**) may be a single integrated or monolithic piece.

[0022] Referring to FIGS. **4** and **6**, the tool housing **102** may enclose or house internal components of the tool **100**, such as the motor **106**, switch assembly **120**, motor control electronics and/or controller **122** (which may be incorporated into the switch assembly **120** or separate), and power source components, such as terminals **124** adapted to operably couple to a battery, such as a rechargeable type battery) that may be received in a battery receiving portion **126** of the tool housing **102**. The tool may also include other components, such as a display for configuring the tool and/or providing operational information about the tool (such as, for example, speed, battery level, etc.), one or more indicators such as light emitting diodes, and other components for operation of the tool, for example. The tool housing **102** may also include a textured grip to improve a user's grip of the tool **100** during use.

[0023] The motor **106** is disposed in and supported in the tool housing **102** and operably coupled to the trigger **108** via the switch assembly **120**. The motor **106** includes a motor shaft **128** (as shown in FIG. **7**) that is operably coupled to the output assembly **104**, as described below. Thus, actuation of the trigger **108** by a user causes the motor **106** to operate and rotate the output assembly **104** in a desired rotational direction, based on a position of the direction selector switch **110**.

[0024] The motor **106** may be a brushless (BLDC) or brushed type motor, or any other suitable motor. A power source can be associated with the tool **100** to provide electric or other forms of power to the tool **100**, such as, for example, hydraulic, or pneumatic, to operate the motor **106**. In an embodiment, the power source (not shown) can be housed in the battery receiving portion **126** of the tool housing **102**, or any other portion of the tool **100**/tool housing **102**. The power source may also be an external component that is not housed by the tool **100**, but that is operatively coupled to the tool **100** through, for example, hoses, wires, or wireless means. In an embodiment, the power source is a removable and rechargeable battery that is adapted to be disposed in the battery receiving portion **126** of the tool housing **102** and electrically coupled to corresponding terminals **124** of the tool **100**.

[0025] In an example, the motor **106** is a brushless DC (BLDC) motor, and the tool **100** includes motor control electronics and/or controller(s) **122** (which may be incorporated into the switch assembly **120** or separate) operably coupled to and adapted to control the motor **106**. For example, referring to FIGS. **4** and **6**, the motor control electronics **122** may include a printed circuit board (PCB) including one or more switching elements disposed thereon. The switching elements may be field effect transistors (FETs), such as, for example, metal-oxide semiconductor field-effect transistors (MOSFETs). In an embodiment, the switching elements may include three high-side switching elements, H**1**, H**2**, and H**3**, and three low-side switching elements, L**1**, L**2**, and L**3**, each being operable in either one of a first or conducting state and a second or non-conducting state. The switching elements are controlled by the PCB to selectively apply power from a power source (e.g., a battery pack) to the motor **106** to achieve desired commutation. By selectively activating particular high-side and low-side switching elements, the motor **106** is operated by having the motor control electronics or controller **122** send an electric current through coils located on a stationary part of the motor **106**, called a stator. The coils cause a magnetic force to be applied to a rotating part of the motor **106**, called a rotor, when electric current runs through the coils. The rotor contains permanent magnets that interact with the magnetic forces caused by windings of the stator. By selectively activating successive combinations of high and low-side switching elements in a particular order, thereby sending a particular order of current through the windings of the stator, the stator creates a rotating magnetic field which interacts with the rotor causing it to rotate, which causes rotation of the motor shaft **128** in a desired direction and at a desired speed, in a well-known manner.

[0026] Referring to FIGS. **4-9**, the output assembly **104** includes an output assembly housing **130** that houses components of the output assembly **104**. The components may include a balancer **132** with a balancer shaft **134** operably coupled to the motor shaft **128**, and a bearing **136** disposed around an exterior surface of the balancer shaft **134** to facilitate rotation of the balancer shaft **134** relative to the output assembly housing **130**. A counterbalance **138** may also be disposed on the exterior surface of the balancer shaft **134** and co-rotate with the balancer **132**. For example, the counterbalance **138** may be coupled to the balancer shaft **134** via a keyed engagement that includes a key **140** disposed in corresponding grooves **142**, **144** of the respective balancer shaft **134** and the counterbalance **138**.

[0027] The output assembly **104** may also include an output shaft **146** disposed in the balancer **132**, a sprag clutch **148** (also known as a one-way bearing, freewheel clutch, cam clutch, or overrunning clutch) disposed on the output shaft **146**, and optionally one or more bearings **150** disposed on the output shaft **146**. An accessory **152**, such as, for example, a backing pad, replaceable polishing pads or discs, or abrasive attachments (such as sanding pads or discs, etc.), or other type of accessory, may be coupled to and end of the output shaft **146**.

[0028] The sprag clutch **148** and optional bearings **150** may be disposed on an exterior surface of the output shaft **146** or otherwise operably coupled to the output shaft **146**. For example, referring to FIGS. **5**, **8**, and **9**, the sprag clutch **148** may be disposed on the output shaft **146** proximal to an end of the output shaft **146** opposite the accessory **152**, and the output shaft **146** retained in the balancer **132**, via the bearings **150** by a retaining ring **154** disposed in a groove in the output shaft **146** between the bearings **150** and the sprag clutch **148**. The bearings **150** may be disposed on the output shaft **146** proximal to an end of the output shaft **146** proximal to the accessory **152**, and the bearings **150** retained on the output shaft **146** and in the balancer **132** by a retaining ring **156** and washer **158**, where the retaining ring **156** is disposed in a groove in an interior surface of the balancer **132**. Thus, the bearings **150** may be sandwiched between the retaining ring **156** and washer **158** and the retaining ring **154**.

[0029] The sprag clutch **148** and optional bearings **150** may also be in contact with the interior surface of the balancer **132**. The sprag clutch **148** allows the output shaft **146** to rotate relative to the balancer **132** when the balancer **132** is rotated (via operation of the motor **106**) in a first rotational direction, and prevents the output shaft **146** from rotating relative to the balancer **132** when the balancer **132** is rotated in a second rotational direction opposite of the first rotational direction. Thus, when the tool **100** is operated in the first direction, the sprag clutch **148** allows free spinning of the output shaft **146**, thus providing operation in a random orbital mode. When the tool **100** is operated in the second direction, the sprag clutch **148** locks or prevents the output shaft **146** from free spinning, thus providing operation in a rotary mode.

[0030] The output assembly **104** may be operably coupled to the motor shaft **128** via one or more other components. For example, a shaft extension **160** may be coupled to an end of the balancer shaft **134** proximal to the motor **106** and operably coupled to the motor shaft **128** via one or more spacers **162**. A bearing **164** may also be disposed on an exterior surface of the shaft extension **160** to facilitate rotation of the shaft extension **160** relative to the output assembly housing **130**. A spring washer **166** may also be disposed between the bearing **164** and the counterbalance **138**. The tool housing **102** may also include a middle housing portion **168** that houses one or more of the components and facilitates coupling of the output assembly housing **130** to the motor housing portion **112**. For example, the output assembly housing **130**, middle housing portion **168**, and motor housing portion **112** may be coupled together via one or more fasteners **170**.

[0031] The supply of power to the motor **106** may be controlled by the trigger **108**. As illustrated, the trigger **108** is disposed in and extends from the handle housing portion **114** proximal to the motor housing portion **112**. The trigger **108** can be actuated by a user to cause power to be supplied from the power source to the motor **106** to drive the motor **106** (and motor shaft **128**) in either one of first and second rotational directions (e.g., clockwise and counterclockwise). The trigger **108** can be biased, such that a user can depress the trigger **108** inwardly, relative to the tool **100**, to cause the tool **100** to operate, and release the trigger **108**, wherein the biased nature of the trigger **108** causes the trigger **108** to move outwardly, relative to the tool **100**, to cease operation of the tool **100**.

[0032] The trigger **108** may also be operably coupled to the switch assembly **120** and motor control electronics **124** to cause power to be supplied from the power source to the motor **106** when the trigger **108** is actuated. Additionally, the trigger **108** and/or switch assembly **120** may also include a variable speed type mechanism. In this regard, actuation of the trigger **108** causes the motor **106** to operate at a faster speed the further the trigger **108** is actuated. However, any suitable trigger **108** or switch assembly **120** can be implemented without departing from the spirit and scope of the present invention.

[0033] The tool **100** may also include a direction selector switch **110** that is actuatable via one or more buttons between first and second positions, to allow a user to select either one of first or second rotational directions (e.g., clockwise and counterclockwise). As illustrated, the direction selector switch **110** is disposed above the trigger **108**, and extends out of the handle housing portion **114**, on opposing sides proximal to the motor housing portion **112**. When the direction selector switch **110** is in the first position, the first rotational direction is selected, and when the direction selector switch **110** is in the second position, the second rotational direction is selected. Selection of the first rotational direction causes the motor **106** to rotate the motor shaft **128** in the first direction, such as clockwise. Similarly, selection of the second rotational direction causes the motor **108** to rotate the motor shaft **128** in the second direction opposite the first direction, such as counterclockwise.

[0034] During operation, when the direction selector switch **110** is in the first position and the trigger **108** is actuated, power is supplied from the power source to the motor **106** to drive the motor **106** (and motor shaft **128**) in the first rotational direction. Rotation of the motor shaft **128** causes rotation of the balancer **132** in the first rotational direction, which rotates the accessory **152** via the output shaft **146**. When the tool **100** is operated in the first rotational direction, the sprag clutch **148** allows free spinning of the output shaft **146** relative to the balancer **132**, providing operation in a random orbital mode.

[0035] Similarly, when the direction selector switch **110** is in the second position and the trigger **108** is actuated, power is supplied from the power source to the motor **106** to drive the motor **106** (and motor shaft **128**) in the second rotational direction. Rotation of the motor shaft **128** causes rotation of the balancer **132** in the second rotational direction, which rotates the accessory **152** via the output shaft **146**. When the tool **100** is operated in the second rotational direction, the sprag clutch **148** locks/prevents free spinning of the output shaft **146** relative to the balancer **132**, thus providing operation in a rotary mode.

[0036] While the sprag clutch **148** is described as allowing free spinning of the output shaft **146** relative to the balancer **132** when the tool **100** is operated in the first rotational direction, and locking/preventing the output shaft **146** from free spinning relative to the balancer **132** when the tool **100** is operated in the second rotational direction, it will be appreciated that operation of the sprag clutch **148** may be reversed. For example, the sprag clutch **148** may allow free spinning of the output shaft **146** relative to the balancer **132** when the tool **100** is operated in the second rotational direction, and lock/prevent the output shaft **146** from free spinning relative to the balancer **132** when the tool **100** is operated in the first rotational direction.

[0037] The tool **100** may also include a nose housing **172** with illuminating elements coupled to the output assembly housing **130** to provide light to the work area. For example, the nose housing **172** may be coupled to the output assembly housing **130** and middle housing **168** via a clip that extends into grooves in the output assembly housing **130** and middle housing **168**. A light circuit board **174** with one or more light emitting diodes (LEDs) **176** may be disposed in the nose housing **172** and on the exterior surface of the output assembly housing **130**, and operably coupled to the power source, such as via the switch assembly **120**. Thus, when the trigger **108** is actuated, power may also be supplied to the light circuit board **174** and LEDs **176** to cause the LEDs **176** to illuminate. The light circuit board **174** may also control the LEDs **176** with a timer. So, for example, when the trigger **108** is released, the LEDs **176** may remain illuminated a predetermined amount of time, for example 5 seconds. Alternately, partial actuation of the trigger **108**, but before the motor **106** is provided power, may cause the LEDs **176** to illuminate. Alternately, a separate switch may also be provided to operate the LEDs. The LEDs **176**, when illuminated, in combination with the nose housing **172**, direct light to illuminate an area of a work surface when the tool **100** is being operated.

[0038] The tool **100** may also include an auxiliary handle **178** for facilitating two-handed operation of the tool **100** to provide better stability during use of the tool **100**. The auxiliary handle **178** may be optionally coupled to one or more handle coupling locations **180** on an exterior surface of the output assembly housing **130**. For example, the output assembly housing **130** may include at least two coupling locations **180** in the form of threaded apertures, and the auxiliary handle **178** may be independently coupled to either coupling location **180** by threadably engaging the auxiliary handle **178** with the threaded apertures.

[0039] In another embodiment, referring to FIG. **10**, the output assembly **104** may include an output shaft **246** (instead of the output shaft **146**) disposed in the balancer **132**, a sprag clutch **248** (instead of the sprag clutch **148**) disposed on the output shaft **246**, and optionally one or more bearings **150** disposed on the output shaft **246**. In this embodiment, the sprag clutch **248** and optional bearings **150** may be disposed on an exterior surface of the output shaft **246** or otherwise operably coupled to the output shaft **246** and sandwiched between the retaining ring **156** and washer **158** and the retaining ring **154**. The sprag clutch **248** and optional bearings **150** may also be in contact with the interior surface of the balancer **132**.

[0040] Like sprag clutch **148**, sprag clutch **248** allows the output shaft **246** to rotate relative to the balancer **132** when the balancer **132** is rotated (via operation of the motor **106**) in a first direction, and prevents the output shaft **246** from rotating relative to the balancer **132** when the balancer **132** is rotated in a second direction. Thus, when the tool **100** is operated in the first direction, the sprag clutch **248** allows free spinning of the output shaft **246**, thus providing operation in a random orbital mode. When the tool **100** is operated in the second direction, the sprag clutch **248** locks/prevents the output shaft **246** from free spinning, thus providing operation in a rotary mode.

[0041] In another embodiment, referring to FIG. **11**, the output assembly **104** may include an output shaft **346** (instead of output shaft **146**) disposed in the balancer **132** and a sprag clutch **348** (instead of sprag clutch **148**) disposed on the output shaft **346**. In this embodiment, the sprag clutch **348** may be disposed on an exterior surface of the output shaft **346** or otherwise operably coupled to the output shaft **346** and sandwiched between the retaining ring **156** and washer **158** and the retaining ring **154**. The sprag clutch **348** may also be in contact with the interior surface of the balancer **132**. Like sprag clutch **148**, sprag clutch **348** allows the output shaft **346** to rotate with respect to the balancer **132** when the balancer **132** is rotated (via operation of the motor **106**) in a first direction, and prevents the output shaft **346** from rotating relative to the balancer **132** when the balancer **132** is rotated in a second direction. Thus, when the tool **100** is operated in the first direction, the sprag clutch **348** allows free spinning of the output shaft **346**, thus providing operation in a random orbital mode. When the tool **100** is operated in the second direction, the sprag clutch **348** locks/prevents the output shaft **346** from free spinning, thus providing operation in a rotary mode.

[0042] As discussed herein, the tool **100** can be an electric tool, such as, for example, a polisher and/or sander. However, it will be appreciated that the tool **100** can alternatively be pneumatically or hydraulically powered.

[0043] As used herein, the term “coupled” and its functional equivalents are not intended to

necessarily be limited to direct, mechanical coupling of two or more components. Instead, the term “coupled” and its functional equivalents are intended to mean any direct or indirect mechanical, electrical, or chemical connection between two or more objects, features, work pieces, and/or environmental matter. “Coupled” is also intended to mean, in some examples, one object being integral with another object. As used herein, the term “a” or “one” may include one or more items unless specifically stated otherwise.
[0044] The matter set forth in the foregoing description and accompanying drawings is offered by way of illustration only and not as a limitation. While particular embodiments have been shown and described, it will be apparent to those skilled in the art that changes and modifications may be made without departing from the broader aspects of the inventors' contribution. The actual scope of the protection sought is intended to be defined in the following claims when viewed in their proper perspective based on the prior art.

Claims

1. An orbital tool, comprising: a tool housing including a motor housing portion with a motor housing longitudinal axis and a handle housing portion with a handle housing longitudinal axis, wherein the handle housing longitudinal axis is disposed at an angle of about 90 degrees to 120 degrees relative to the motor housing longitudinal axis; a motor disposed in the motor housing portion and including a motor shaft, wherein the motor is adapted to selectively cause the motor shaft to rotate in either of first and second rotational directions; a balancer operably coupled to the motor shaft; an output shaft disposed in the balancer; and a sprag clutch disposed in the balancer and operably coupled to the output shaft, wherein the sprag clutch is adapted to allow rotation of the output shaft relative to the balancer when the motor shaft is rotated in the first rotational direction, and prevent rotation of the output shaft relative to the balancer when the motor shaft is rotated in the second rotational direction.
2. The orbital tool of claim 1, further comprising a bearing disposed in the balancer and operably coupled to the output shaft.
3. The orbital tool of claim 1, further comprising a counterbalance coupled to an exterior surface of the balancer.
4. The orbital tool of claim 3, wherein the counterbalance is coupled to the balancer via a keyed engagement.
5. The orbital tool of claim 4, wherein the keyed engagement includes a key disposed in corresponding grooves of the balancer and the counterbalance.
6. The orbital tool of claim 1, further comprising an auxiliary handle adapted to be coupled to the tool.
7. The orbital tool of claim 1, further comprising a backing pad coupled to the output shaft.
8. An orbital tool, comprising: a tool housing; a nose housing including an illumination element and coupled to the tool housing; a motor disposed in the tool housing and including a motor shaft, wherein the motor is adapted to selectively cause the motor shaft to rotate in either of first and second rotational directions; a balancer operably coupled to the motor shaft; an output shaft disposed in the balancer; and a sprag clutch disposed in the balancer and operably coupled to the output shaft, wherein the sprag clutch is adapted to allow rotation of the output shaft relative to the balancer when the motor shaft is rotated in the first rotational direction, and prevent rotation of the output shaft relative to the balancer when the motor shaft is rotated in the second rotational direction.
9. The orbital tool of claim 8, wherein the tool housing includes a motor housing portion with a motor housing longitudinal axis and a handle housing portion with a handle housing longitudinal axis, wherein the handle housing longitudinal axis is disposed at an angle of about 90 degrees to 120 degrees relative to the motor housing longitudinal axis.

10. The orbital tool of claim 8, further comprising a counterbalance coupled to an exterior surface of the balancer.
11. The orbital tool of claim 10, wherein the counterbalance is coupled to the balancer via a keyed engagement.
12. The orbital tool of claim 11, wherein the keyed engagement includes a key disposed in respective corresponding grooves of the balancer and the counterbalance.
13. The orbital tool of claim 8, further comprising an auxiliary handle adapted to be coupled to the tool.
14. The orbital tool of claim 8, further comprising a backing pad coupled to the output shaft.
15. An orbital tool, comprising: a housing including a handle housing portion; a motor disposed in the housing and including a motor shaft, wherein the motor is adapted to cause the motor shaft to rotate in either of first and second rotational directions; a balancer operably coupled to the motor shaft; a counterbalance coupled to an exterior surface of the balancer; an output shaft disposed in the balancer; and a sprag clutch disposed in balancer and operably coupled to the output shaft, wherein the sprag clutch is adapted to allow rotation of the output shaft relative to the balancer when the motor shaft is rotated in the first rotational direction, and prevent rotation of the output shaft relative to the balancer when the motor shaft is rotated in the second rotational direction.
16. The orbital tool of claim 15, further comprising a bearing disposed in the balancer and operably coupled to the output shaft.
17. The orbital tool of claim 15, further comprising an auxiliary handle adapted to be coupled to the tool.
18. The orbital tool of claim 15, further comprising a backing pad coupled to the output shaft.
19. The orbital tool of claim 15, further comprising a trigger operably coupled to the motor, wherein actuation of the trigger is adapted to cause power to be supplied to the motor.
20. The orbital tool of claim 15, further comprising a direction selector switch operably coupled to the motor and actuatable to select either of the first and second rotational directions.